

# Learning and the emergence of nonlinearity in financial markets

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<sup>1</sup>The views in this presentation are those of the authors and do not represent those of the Federal Reserve System or Board of Governors.

# Introduction

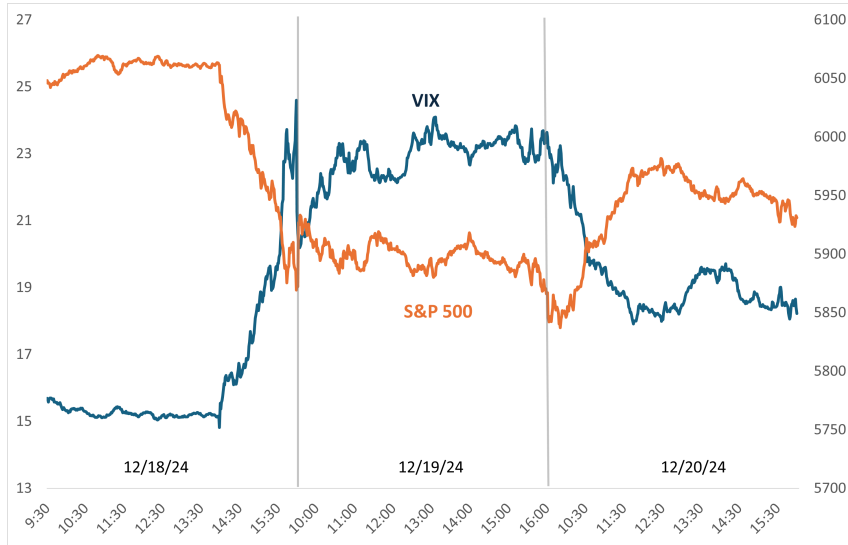
- ▶ Stock market returns are highly nonlinear/non-gaussian
  - ▶ Conditionally and unconditionally negatively skewed
  - ▶ Heavy-tailed (crashes)
  - ▶ Volatility time-varying, skewed, and has long memory
  - ▶ Volatility strongly related to market level
- ▶ **This paper:** all of these nonlinear patterns emerge naturally in a simple learning framework

# Facts we are targeting

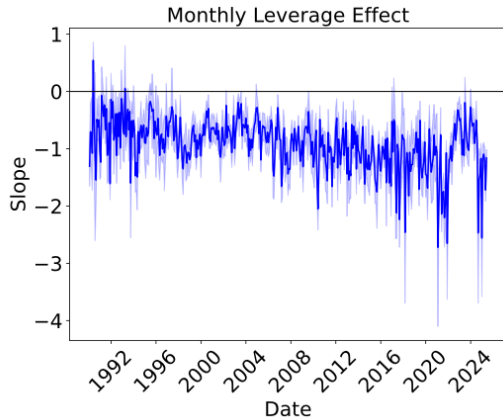
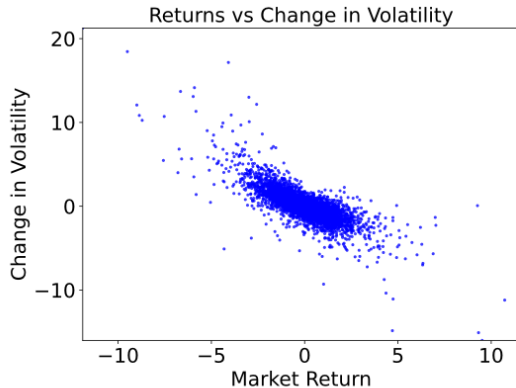
## 1. The leverage effect:

- ▶ Volatility rises as prices fall
- ▶ 70% daily, 80% 15-minute correlation between S&P 500 returns and volatility changes
- ▶ Creates a long left tail in returns

## Facts we are targeting



## Facts we are targeting



# Facts we are targeting

Standard theoretical explanations for the **leverage effect**

## 1. Financial leverage

- ▶ Can't be just that, as leverage effect is there even for all-equity firms
- ▶ Too small quantitatively

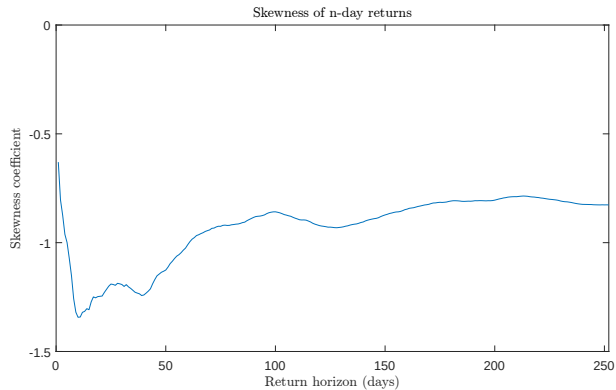
## 2. **Volatility drives risk premia** and hence prices: French, Schwert, and Stambaugh (1987); Campbell and Hentschel (1992); Bollerslev and Todorov (2023)

- ▶ Would require strong time-series relation between volatility and risk premia

## 3. **Learning** (about model parameters or state of the world): Johannes, Lochstoer, and Mou (2016); Weitzman (2007); Veldkamp and Orlik (2024); Veronesi (1999); David and Veronesi (2013) ; Kozlowski, Veldkamp, and Venkateswaran (2019); Collin-Dufresne, Johannes, and Lochstoer (2016)

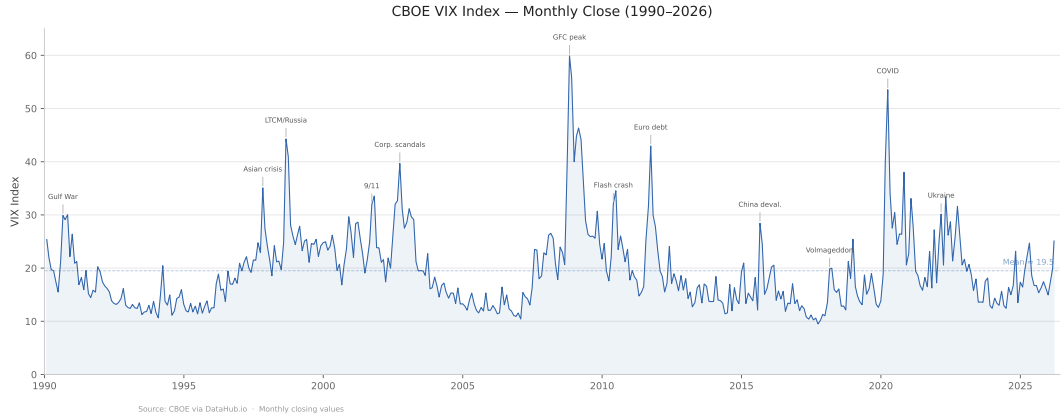
- ▶ Many **specialized** learning models, some of which *can* generate some facts but miss on other moments

## Facts we are targeting



### 2. Skewness of returns over different holding periods

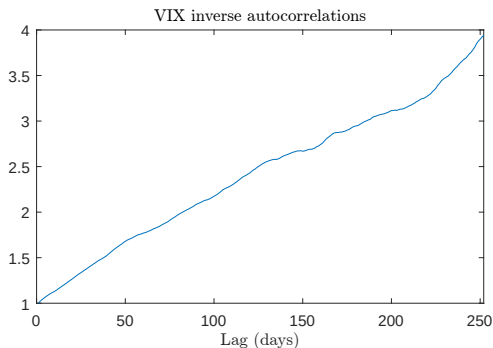
# Facts we are targeting



## 3. Slow decay of volatility



## Facts we are targeting



### 3. Slow decay of volatility

- ▶ Inverse autocorrelations of volatility ( $1/\text{autocorr}(j)$ ) close to linear (not exponential as in ARMA model)
- ▶ “Long memory”, typically modeled in reduced form

# What we do

- ▶ Introduce a general **filtering** model, where investors have arbitrary belief distribution about fundamentals, updated over time
- ▶ **Key theoretical result:**
  - ▶ Recursive description of how the entire posterior distribution updates when news arrives
  - ▶ Sensitivity of *mean* to news  $\propto$  conditional *variance*
  - ▶ Sensitivity of *variance* to news  $\propto$  conditional *third moment*
  - ▶ Sensitivity of *third moment* to news  $\propto$  conditional *fourth moment* ...
- ▶ Characterize the **joint** dynamics of all moments of beliefs about the fundamental

# What we do

Derive implications when agents learn about the fundamental value of the stock market and prices are the mean of the beliefs distribution:

1. Skewness in the distribution of fundamentals is necessary and sufficient to generate the **leverage effect**
2. **Skewness** increases with horizon
3. Volatility displays **long memory**
4. All these nonlinearities emerge **jointly** and are explained **quantitatively**

Bottom line: a simple filtering framework with non-normal prior can naturally generate many of the observed empirical properties of prices

# Theoretical setup

# Prices

- ▶ Conditional on an information set,  $\mathcal{I}_t$ , prices are

$$P_t(\mathcal{I}_t) = E \left[ \int_{s=0}^{\infty} \frac{D_{t+s} M_{t+s}}{M_t} ds \mid \mathcal{I}_t \right]$$

- ▶  $D$  cash flows,  $M$  SDF process
- ▶ Restriction:  $M$  is exogenous to  $\mathcal{I}_t$
- ▶ **Fundamental value** is

$$X_t \equiv P_t(\theta_t)$$

- ▶  $X_t$  process essentially unrestricted

# Information

- ▶ Investors observe a noisy signal  $Y$ , and use it to try to learn the fundamental value:
- ▶ Law of iterated expectations:

$$P_t(Y^t) = E[X_t | Y^t]$$

- ▶  $X_t$  contains all risk adjustments  $\Rightarrow$  filtering is risk-adjusted
- ▶ Stock prices are a geometric process. We need one elision:

$$p_t \equiv E[x_t | Y^t]$$

$$\text{where } x_t \equiv \log X_t$$

log price is expectation of log fundamentals (note that quantitative example doesn't use this)

# Diffusive information

- ▶ Information is diffusive (Gaussian):

$$dY_t = x_t dt + \sigma_{Y,t} dW_t$$

- ▶ Not the only info structure: could observe discrete signals, signals about functions of  $x$
- ▶ Seems natural:  $x_t$  is what you'd pay with full info

# The whole model

- ▶ The entire model is

$$\begin{aligned} p_t &\equiv E[x_t | Y^t] \\ dY_t &= x_t dt + \sigma_{Y,t} dW_t \end{aligned}$$

- ▶ If  $x_t$  is linear and Gaussian, can use Kalman filter
- ▶ What if it's not? Generally difficult – closed forms rarely available



# Solution

# Filtering

- ▶ Agents' state is their subjective distribution for  $x_t$ . Moments  $\kappa_{1,t}, \kappa_{2,t}, \dots$

**Proposition:** The dynamics of the first two moments of beliefs are

$$\underbrace{d\kappa_{1,t}}_{\text{Expectation (=prices)}} = \kappa_{2,t} \sigma_Y^{-2} \underbrace{(dY_t - E_t[x_t] dt)}_{\text{Surprise in signal}} + E_t[dx_t]$$
$$\underbrace{d\kappa_{2,t}}_{\text{Uncertainty}} = \kappa_{3,t} \sigma_Y^{-2} \underbrace{(dY_t - E_t[x_t] dt)}_{\text{Surprise in signal}} + \underbrace{E_t[d(x_t^2)]}_{\text{Fund. shocks}} - \underbrace{\kappa_{2,t}^2 \sigma_Y^{-2} dt}_{\text{Info. flow}}$$

- ▶ Sensitivity of prices to signals: uncertainty times signal precision
- ▶ Sensitivity of uncertainty to signals: third moment times precision
- ▶ Note: implies prices are a diffusion

# Filtering

**Proposition.** The dynamics of the first two moments of beliefs are

$$\underbrace{d\kappa_{1,t}}_{\text{Expectation (=prices)}} = \underbrace{\kappa_{2,t}\sigma_Y^{-2}}_{\text{Gain}} \underbrace{(dY_t - E_t[x_t] dt)}_{\text{Surprise in signal}} + E_t[dx_t]$$

$$\underbrace{d\kappa_{2,t}}_{\text{Uncertainty}} = \underbrace{\kappa_{3,t}\sigma_Y^{-2}}_{\text{Gain}} \underbrace{(dY_t - E_t[x_t] dt)}_{\text{Surprise in signal}} + E_t[d(x_t^2)] - \underbrace{\kappa_{2,t}^2\sigma_Y^{-2} dt}_{\text{Info. flow}}$$

► Intuition: local Kalman filter

- Price gain:  $\kappa_{2,t}\sigma_Y^{-2} = \frac{\text{cov}(x_t, dY_t)}{\text{var}(dY_t)}$  local regression coefficient
- Uncertainty gain:  $\kappa_{3,t}\sigma_Y^{-2} = \frac{\text{cov}((x - E[x])^2, dY_t)}{\text{var}(dY_t)}$  same
- $\kappa_{2,t}^2\sigma_Y^{-2} dt$  measures information flow

# General result

## Theorem

*For all  $n$  such that the cumulants exist,*

$$d\kappa_{n,t} = \frac{\kappa_{n+1,t}}{\sigma_{Y,t}^2} (dY_t - E_t[x_t] dt) - \frac{1}{\sigma_{Y,t}^2} \sum_{j=2}^n \alpha_j^{(n)} \kappa_{j,t} \kappa_{n-j+2,t} dt + E_t \left[ d \left( x_t^k \right) \right]$$

- Fully characterizes dynamics of the distribution

# Predictions

# 1. Volatility is proportional to uncertainty

- ▶ **Prices** are posterior mean  $\Rightarrow$

$$dp_t = d\kappa_{1,t} = \frac{\kappa_{2,t}}{\sigma_{Y,t}^2} (dY_t - E_t[x_t] dt) + E_t[dx_t]$$

- ▶ Therefore, they are an Ito diffusion
- ▶ Instantaneous return volatility:

$$\text{std}(dp_t) = \frac{\kappa_{2,t}}{\sigma_{Y,t}} dt^{1/2}$$

- ▶ **Volatility** is proportional to uncertainty (though there's a wedge –  $\sigma_{Y,t}$ )

## 2. Leverage effect $\iff$ Belief skewness

$$d[\text{std}(dp_t)] = \sigma_{Y,t}^{-1} \left( \frac{\kappa_{3,t}}{\kappa_{2,t}} dp_t + E_t[d(x_t^2)] \right) - \frac{1}{\sigma_{Y,t}} [\text{std}(dp_t)]^2 dt$$

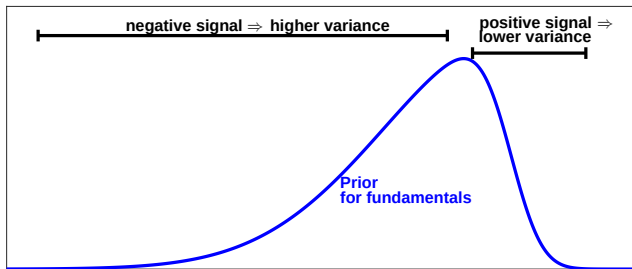
- The instantaneous **leverage effect** is

$$\frac{\text{cov}(dp_t, d[\text{std}(dp_t) dt^{-1/2}])}{\text{var}(dp_t)} = \frac{\kappa_{3,t}}{\sigma_{Y,t}^2 \kappa_{2,t}}$$

- Leverage effect  $\iff$  **negative skewness in beliefs** ( $\kappa_{3,t} < 0$ )
- Note stochastic vol. also implies excess kurtosis in dp

## 2. Leverage effect $\iff$ Belief skewness

- ▶ Leverage effect  $\iff$  **negative skewness in beliefs** ( $\kappa_{3,t} < 0$ )
  - ▶ Intuition: if the distribution is left skewed, negative shock brings us to the wider part of the distribution, and uncertainty goes up
  - ▶ Intuition is obvious; what's surprising is that the third moment is the sufficient statistic in **all** cases (no matter what the belief distribution is)





### 3. Long memory in volatility

$$d[\text{std}(dp_t)] = \sigma_{Y,t}^{-1} \left( \frac{\kappa_{3,t}}{\kappa_{2,t}} dp_t + E_t[d(x_t^2)] \right) - \frac{1}{\sigma_{Y,t}} [\text{std}(dp_t)]^2 dt$$

- ▶ Rate of learning proportional to squared uncertainty
- ▶ Uncertainty spikes, initially reverts fast, then slows
- ▶ Yields long memory as in Ding, Engle, and Granger (1993) (also Mandelbrot)

## 4. Skewness patterns of returns

- ▶ Combining  $dp$  and  $d\kappa_2$ ,

$$\text{skew}_t(dp_t) \approx 3 \frac{\kappa_{3,t}}{\kappa_{2,t}} \sigma_{Y,t}^{-1} dt^{1/2}$$

- ▶ Returns are skewed because volatility rises when prices fall
  - ▶ Still locally normal: as  $dt \rightarrow 0$ ,  $\text{skew}(dp) \rightarrow 0$
- ▶ Return skewness is fundamentally driven by belief skewness ( $\kappa_{3,t}$ )
  - ▶ Second moment also matters, since the second moment determines how strongly prices move as volatility changes

## Example: linear Gaussian process

- ▶ If  $x_t$  driven by a linear Gaussian process, can apply Kalman filter
- ▶  $\mu_{3,t} = 0$ ,  $E_t \left[ (dx_t)^2 \right] = \sigma_X^2$  constant

$$dp_t = E_t[dx_t] + \frac{\kappa_{2,t}}{\sigma_{Y,t}^2} (dY_t - E_t[x_t] dt)$$
$$d\kappa_{2,t} \rightarrow \text{constant}$$

- ▶ No leverage effect, volatility deterministic, no skewness, etc.
- ▶ **No other distribution** has cumulants equal to zero after some cutoff
  - ▶ Can't just shut down higher-order variation

## Example: Markov switching model (Veronesi)

- ▶ Suppose  $x$  switches between low and high values according to a Markov process
- ▶ Current state is probability of being in the high regime,  $\pi_t$

$$\kappa_{2,t} = \pi_t(1 - \pi_t)$$

$$\kappa_{3,t} = (1 - 2\pi_t) \times \kappa_{2,t}$$

- ▶ Implication: when times are bad ( $\pi_t$  low), skewness changes sign

# Quantitative Example

# Calibration

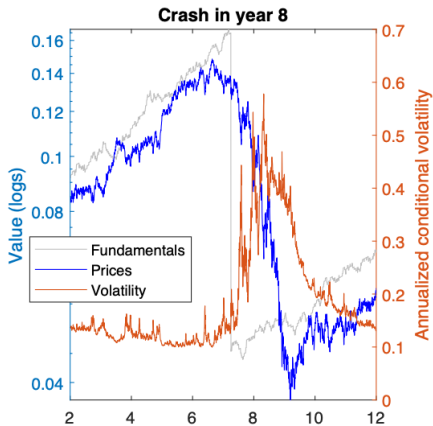
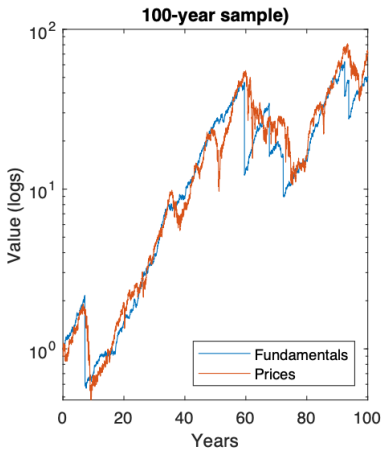
- Fundamentals follow a random walk with jumps:

$$dx_t = (\phi\lambda + g)dt + \underbrace{\sigma_X dB_t}_{\text{Diffusive risk}} - \underbrace{J_t dN_t}_{\text{Jumps}}$$

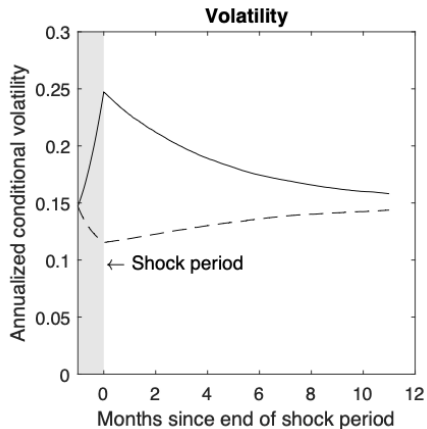
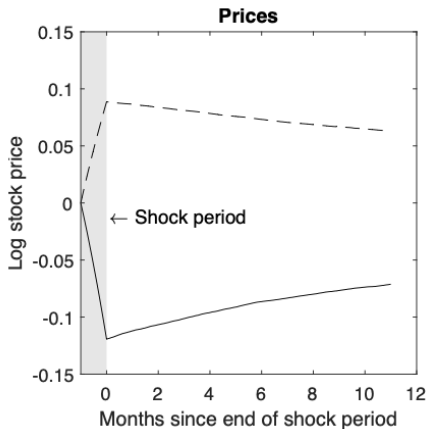
$J_t$  distr. as in Barro and Jin (2011)

- Selected parameter values to fit stock market moments:
  - Jump probability = 0.037 (per year)
  - Noise in signals:  $\sigma_Y = 6.5$  (daily)
  - Diffusive volatility:  $\sigma_X = 0.065$  (per year)

# 100 year simulation

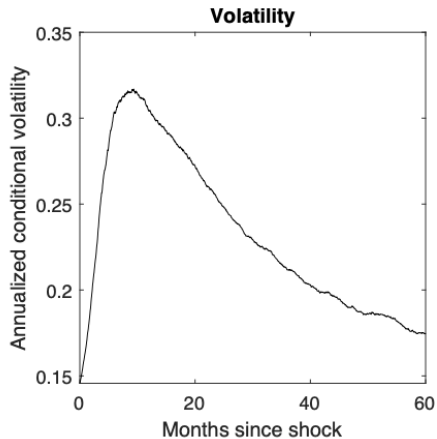
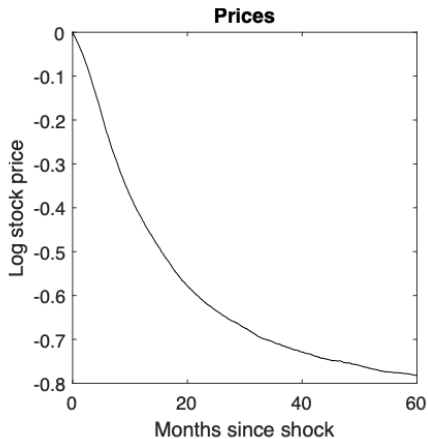


## Response to negative noise (1 in 10 year one-month shock)

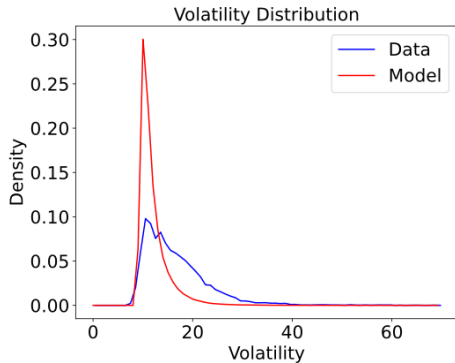
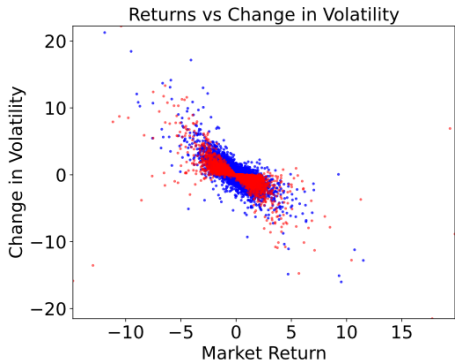




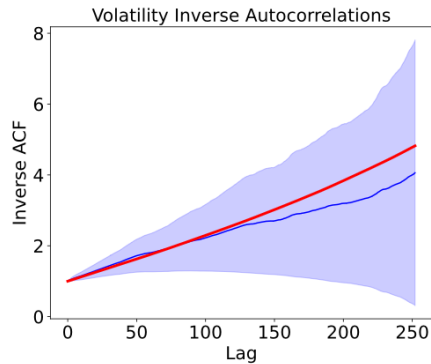
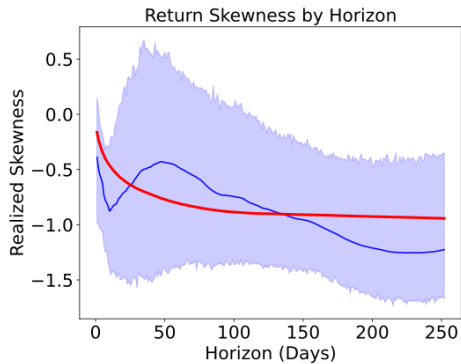
## Response to jump in fundamentals



# Model vs. data



## Model vs. data



## Model vs. data: moments

| Moment         | Stock returns |       | Volatility level |       | Volatility change |       |
|----------------|---------------|-------|------------------|-------|-------------------|-------|
|                | Data          | Model | Data             | Model | Data              | Model |
| Std. dev.      | 1.14          | 0.98  | 6.73             | 6.72  | 1.43              | 0.88  |
| Skewness       | -0.06         | 0.02  | 2.16             | 4.25  | 1.37              | 0.15  |
| Kurtosis       | 19.54         | 12.14 | 11.38            | 33.35 | 30.36             | 75.88 |
| Corr. w/ $R_t$ |               |       |                  |       | -0.78             | -0.76 |

# Nonparametric tests

## Overidentifying restrictions

- ▶ Relationships in the dynamics of **moments of beliefs** translate into relationship between **moments of prices**:

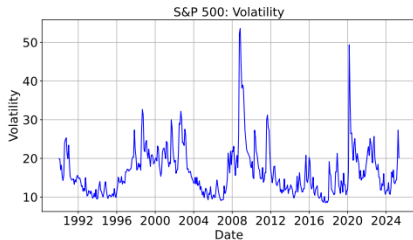
$$d[\text{std}_t(dp_t)] = \textcolor{red}{1} \left( \frac{1}{3} \Delta t^{-1/2} \right) \text{skew}_t(p_{t+\Delta t}) dp_t - \frac{\textcolor{red}{1}}{\textcolor{red}{\sigma_{Y,t}}} \text{vol}_t^2 dt \\ + E_t[d[x]_t] + 2\text{Cov}_t(x_{t-}, dx_t)$$

- ▶ First coefficient restricted to 1
  - ▶ Follows from returns and volatility driven by single diffusive shock
- ▶ Second coefficients measures signal precision
- ▶ Absorb drift into the constant – valid if  $x$  has independent increments

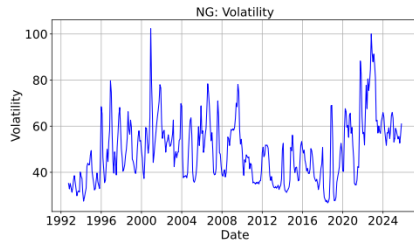
## Overidentifying restrictions

$$\begin{aligned} d[\text{std}_t(dp_t)] &= \textcolor{red}{1} \left( \frac{1}{3} \Delta t^{-1/2} \right) \text{skew}_t(p_{t+\Delta t}) dp_t - \frac{\textcolor{red}{1}}{\textcolor{red}{\sigma}_{Y,t}} \text{vol}_t^2 dt \\ &\quad + E_t[d[x]_t] + 2\text{Cov}_t(x_{t-}, dx_t) \end{aligned}$$

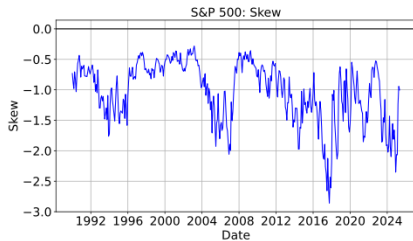
- ▶ We run this for the S&P 500 and natural gas futures
  - ▶ Natural gas because its skewness is typically positive, vs negative for S&P
- ▶ Forecast conditional moments with option-implied moments



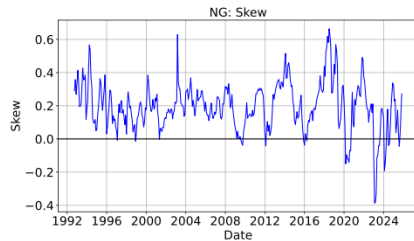
(a) Volatility (S&P 500)



(b) Volatility (natural gas)



(c) Skew (S&P 500)



(d) Skew (natural gas)



# Overidentifying restriction

|                                      | S&P 500                |                       |                        | Natural Gas            |                        |                        |
|--------------------------------------|------------------------|-----------------------|------------------------|------------------------|------------------------|------------------------|
|                                      | (1)                    | (2)                   | (3)                    | (4)                    | (5)                    | (6)                    |
| $Vol_{t-1}^2$                        | -0.7449***<br>[0.1605] | -0.7611<br>[0.5096]   | -0.5712***<br>[0.1355] | -0.1496***<br>[0.0416] | -0.6816***<br>[0.2556] | -0.1497***<br>[0.0415] |
| $\frac{1}{3\sqrt{21}}skew_{t-1}dp_t$ | 1.0078***<br>[0.0265]  | 1.0077***<br>[0.0265] | 0.2784***<br>[0.0410]  | 1.1333***<br>[0.0625]  | 1.1303***<br>[0.0624]  | 1.1140***<br>[0.0831]  |
| $Vol_{t-1}$                          |                        | 0.0005<br>[0.0124]    |                        |                        | 0.0389**<br>[0.0165]   |                        |
| $dp_t$                               |                        |                       | -0.0477***<br>[0.0028] |                        |                        | 0.0005<br>[0.0012]     |
| $R^2$                                | 0.5489                 | 0.5489                | 0.6306                 | 0.1835                 | 0.1856                 | 0.1836                 |

# Estimate of noise and fundamental uncertainty

- ▶ The coefficients here give an estimate of  $\sigma_Y$
- ▶ Coefficient on  $VIX_{t-1}^2$  implies 95% CI for  $\sigma_Y$  is  $[0.94, 2.36]$ 
  - ▶ SD of noise in signals in log points (should be large)
- ▶ Recall  $VIX = \kappa_2 / \sigma_Y$
- ▶ Given  $VIX$ ,  $\sigma_Y$ ,  $\kappa_2^{1/2} \in [10.4, 16.5]$  : estimate of fundamental uncertainty

# Conclusions

- ▶ Main filtering result: describe the evolution of (generic) beliefs over time as news arrives
  - ▶ Main restriction is the structure of information
  - ▶ Allows to get analytical results outside the small space of conjugate priors
- ▶ Many applications to understanding beliefs in financial markets and macroeconomic beliefs
  - ▶ Negative skewness in beliefs sufficient to generate wide range of empirical facts, including leverage effect
  - ▶ Filtering is enough to create quantitatively realistic nonlinearity in return dynamics
  - ▶ Restrictions implied by beliefs dynamics for price dynamics can be tested in the data

# Conclusions

- ▶ Other applications when belief updating is central but underlying process is not linear and Gaussian
  - ▶ Dynamics of inflation beliefs (Dew-Becker, Giglio and Molavi 2026)
  - ▶ Understanding the rate of learning (Christiano, Eichenbaum and Johannsen 2024)
  - ▶ Empirical tests of Bayesian learning